A11y Best Practices

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LXT's 10 A11y Best Practices

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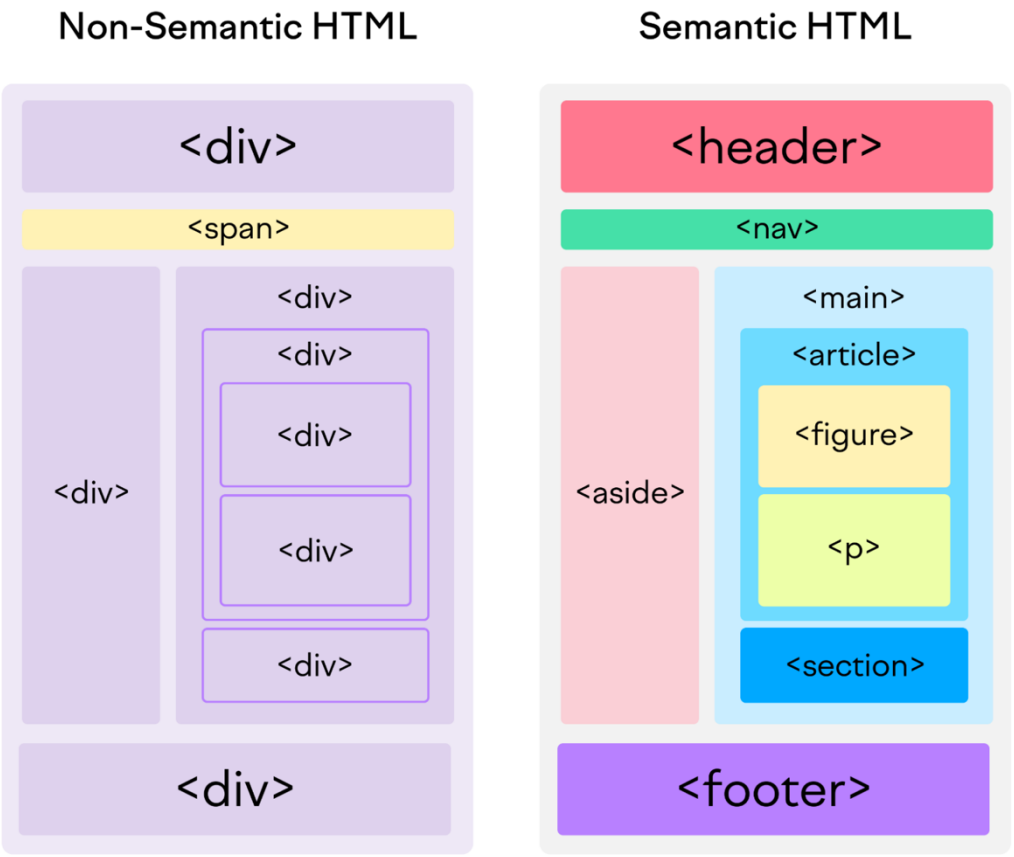
A11y – Best Practices

1. Semantic HTML & Structure
2. Keyboard A11y & Navigable
3. Color & Contrast
4. Text Alts
5. Media
6. Tables
7. Forms
8. Font Size
9. Tap Targets
10. Language

Agenda

Semantic HTML



#1 Semantic

Semantic HTML

```
<div class="wrapper">
  <aside class="sidebar">
    <app-sidebar />
  </aside>

  <main class="main-panel">
    <app-navbar />

    <article class="content">
      <router-outlet />
    </article>

    <app-footer />
  </main>
</div>
```

#1 Semantic

Keyboard A11y & Navigable

- WCAG 2.1, 2.4
- navigation with keyboard etc. must be possible
- use correct focus and tab order [tabindex]
  - avoid [tabindex] > 0
  - can only be "0" (adds focus border/outline) or "-1" (avoid focus)
- focus must be visible and recognizeable
  - use ":focus" to show outline (like ":hover"), avoid "outline: none"
  - must be at least 2px, should be at least 3px (hint: color contrast!)
- use skip links to skip navigations
- a vs. button

Keyboard A11y & Navigable

```
<a routerLink="/home">
  <span class="simple-text logo-mini logo-image-small">
    <img ngSrc="/assets/angular_gradient_68x68.png" width="34"
height="34" alt="Angular Logo Small" />
  </span>

  <span class="simple-text logo-normal">NG a11y</span>
</a>
```

2 Keyboard

Color & Contrast

- WCAG 1.4.1, 1.4.3, 1.4.6
- avoid color as the only visual means of
  - conveying information, indicating an action, prompting a response or distinguishing a visual element
  - links with text-decoration: underline
- use a contrast ratio of at least 4.5:1 for text & images of text (3:1 for headings)
  - https://www.whocanuse.com/

Text Alts

- WCAG 1.1

- use descriptive text [alt] for images & icons
  - can be alt="" (no description / just decorative)
  - icons/image without purpose [aria-hidden]

Text Alts

```
<img
  class="full-width"
  width="2596"
  height="1890"
  ngSrc="/assets/img/photos/grimming_from_glaeserkoppe.jpg"
  alt="Photo showing Grimming from the Gläserkoppe"
  priority
/>
```

Media (Audio & Video)

- WCAG 1.2
- audio-only content
  - text transcript 1.2.1
  - and captions for prerecorded 1.2.2
- video-only content
  - alt text or alternative content 1.2.1
- prerecorded video content
  - text transcript or audio description 1.2.3
- AA
  - captions for live audio content 1.2.4
  - audio description for prerecorded video content 1.2.5

Tables

- WCAG 1.3.1
- use header and footer, use col or row
  - <theader> & <tbody>
  - <th scope="col"> or <th scope="row">
  - if not using native <table> use ARIA table, row, columnheader, rowheader
- use caption & summary
  - <caption> identifies topic and is useful in most situations
  - <summary> provides orientation or navigation hints

Tables

```
<table>
  <caption style="text-align: left">
    Availability of holiday accommodation<br />
    <span style="font-size: 0.75em; display: block">
      Column one has the location and size of accommodation, other
columns show the type and number of properties available
    </span>
  </caption>
  <thead>
    <tr>
      <td></td>
      <th id="stud" scope="col">Studio</th>
      <th id="apt" scope="col"><abbr title="Apartment">Apt</abbr></th>
      <th id="chal" scope="col">Chalet</th>
      <th id="villa" scope="col">Villa</th>
    </tr>
  </thead>
  <tbody>
    [...]
  </tbody>
  <summary>This is an example of an HTML table</summary>
</table>
```

Tables

```
<p id="tableDescription">
Column one has the location and size of accommodation, other
columns show the type and number of properties available.
</p>
<table aria-describedby="tableDescription">
```

Forms

- WCAG 1.3.1, 3.3
- use form elements lables & descriptions
  - <label for="!!!">
  - grouping with <fieldset> & <legend>
  - [aria-describedby] & [aria-requried]
- use good feedback
  - recognizebale, understandable & helping
  - [aria-invalid] & [aria-live]

Forms

```
<label for="from">From (*)</label>
<input
  autocomplete="false"
  type="text"
  name="from"
  id="from"
  class="form-control"
  required
  [minlength]="minLength"
  [maxlength]="maxLength"
  [pattern]="pattern"
  [attr.aria-invalid]="
    flightSearchForm.controls['from']?.dirty && flightSearchForm.controls['from'].invalid
  "
  [attr.aria-describedby]="
    flightSearchForm.controls['from']?.dirty &&
    flightSearchForm.controls['from'].errors
      ? 'from_errors'
      : null
  "
  [(ngModel)]="from"
/>
@if (flightSearchForm.controls['from'] && flightSearchForm.controls['from'].dirty) {
  <app-flight-validation-errors
    id="from_errors"
    [errors]="flightSearchForm.controls['from'].errors"
    fieldLabel="From"
  />
}
```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Font Size

- WCAG 1.4.4
- text can be resized up to 200%
  - without loss of content or functionality
  - except for captions and images of text
- html { font-size: 100% } (and then rem)
- use rem not px (browsers make it work)

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Tap Targets

- WCAG 2.5.5, 2.5.8

- size of the target for pointer inputs is at least 24x24 (better 44x44) CSS pixels

- use padding if element (e.g. icon) is to small

Accessible Language

- WCAG 3.1
- use [lang]
- make text readable & understandable
  - unusual words
  - abbreviations
  - reading level (complexity)
  - pronunciation (dialect)

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Recap

Questions?

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